

CM0081 Formal Languages and Automata

§ 4.2 Closure Properties of Regular Languages

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Preliminaries

Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook (Hopcroft, Motwani and Ullman 2007).
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A , that is, the set of its subsets, is denoted by $\mathcal{P}A$.

Introduction

Let L and L' be regular languages. The following languages are regular:

$L \cup L'$	(union)
$L \cap L'$	(intersection)
$\overline{L}$	(complement)
$L - L'$	(difference)
L^R	(reversal)
L^*	(closure)
$L \cdot L'$	(concatenation)
$h(L)$	(homomorphism)
$h^{-1}(L)$	(inverse homomorphism)

Closure Under Union

Theorem 4.4

If L and L' are regular languages, then so is $L \cup L'$.

Closure Under Union

Theorem 4.4

If L and L' are regular languages, then so is $L \cup L'$.

Proof

(Using regular expressions)

Closure Under Complementation

Definition

Let L be a language over alphabet Σ . The **complement** of L is defined by

$$\overline{L} := \Sigma^* - L.$$

Closure Under Complementation

Theorem 4.5

If L is a regular language, then so is $\overline{L}$.

Closure Under Complementation

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Proof

Let $A = (Q, \Sigma, \delta, q_0, F)$ be a DFA that accepts L . Then $B = (Q, \Sigma, \delta, q_0, Q - F)$ is a DFA that accepts $\overline{L}$. ■

Closure Under Complementation

Question

“Do you see how to take a regular expression and change it into one that defines the complement language?” (Hopcroft, Motwani and Ullman 2007, p. 136)

Closure Under Complementation

Observation

Using the closure properties we can prove that a language is not regular.

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Example

Given that

$$L_{=} = \{ w \in \{0, 1\}^* \mid w \text{ has an equal numbers of 0's and 1's} \}$$

is a language not regular. Prove that

$$L_{\neq} = \{ w \in \{0, 1\}^* \mid w \text{ has an unequal numbers of 0's and 1's} \}$$

is a language not regular.

Closure Under Complementation

Observation

Using the closure properties we can prove that a language is not regular.

Example

Given that

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is a language not regular.

Proof

Whiteboard.

Product Construction

Construction

Let A_L , A_M and A be DFAs given by

$$\begin{aligned}A_L &= (Q_L, \Sigma, \delta_L, q_L, F_L), \\A_M &= (Q_M, \Sigma, \delta_M, q_M, F_M), \\A &= (Q_L \times Q_M, \Sigma, \delta, (q_L, q_M), F_L \times F_M),\end{aligned}$$

where

$$\begin{aligned}\delta &: (Q_L \times Q_M) \times \Sigma \rightarrow Q_L \times Q_M \\ \delta((p, q), a) &= (\delta_L(p, a), \delta_M(q, a)).\end{aligned}$$

Product Construction

Theorem (Exercise 4.2.15)

For all $w \in \Sigma^*$, $\hat{\delta}((q_L, q_M), w) = (\hat{\delta}_L(q_L, w), \hat{\delta}_M(q_M, w))$.

Product Construction

Theorem (Exercise 4.2.15)

For all $w \in \Sigma^*$, $\hat{\delta}((q_L, q_M), w) = (\hat{\delta}_L(q_L, w), \hat{\delta}_M(q_M, w))$.

Proof by induction on w

1. Basis step

$$\begin{aligned}\hat{\delta}((q_L, q_M), \varepsilon) &= (q_L, q_M) && \text{(def. of } \hat{\delta}) \\ &= (\hat{\delta}_L(q_L, \varepsilon), \hat{\delta}_M(q_M, \varepsilon)) && \text{(def. of } \hat{\delta}_L \text{ and } \hat{\delta}_M)\end{aligned}$$

Product Construction

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2. Inductive step

$$\begin{aligned}\hat{\delta}((q_L, q_M), xa) &= \delta(\hat{\delta}((q_L, q_M), x), a) && \text{(def. of } \hat{\delta}) \\ &= \delta((\hat{\delta}_L(q_L, x), \hat{\delta}_M(q_M, x)), a) && \text{(by IH)} \\ &= (\delta_L(\hat{\delta}_L(q_L, x), a), \delta_M(\hat{\delta}_M(q_M, x), a)) && \text{(def. of } \delta) \\ &= (\hat{\delta}_L(q_L, xa), \hat{\delta}_M(q_M, xa)) && \text{(def. of } \hat{\delta}_L \text{ and } \hat{\delta}_M)\end{aligned}$$

Closure Under Intersection

Theorem 4.8

If L and L' are regular languages, then so is $L \cap L'$.

Closure Under Intersection

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If L and L' are regular languages, then so is $L \cap L'$.

Proof

Let A_L and $A_{L'}$ be DFAs accepting L and L' . The product construction of A_L and $A_{L'}$ accepts $L \cap L'$. ■

Closure Under Intersection

Theorem 4.8

If L and L' are regular languages, then so is $L \cap L'$.

Proof

Let A_L and $A_{L'}$ be DFAs accepting L and L' . The product construction of A_L and $A_{L'}$ accepts $L \cap L'$. ■

Different proof

The regular languages are closure under union and complement, and

$$L \cap L' = \overline{\overline{L} \cup \overline{L'}}.$$

■

Closure Under Reversal

Definition

Let $w = a_1 a_2 \cdots a_n$ be a word. The **reversal** of w is defined by

$$w^R := a_n a_{n-1} \cdots a_1.$$

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Definition

Let L be a language on alphabet Σ . The **reversal** of L is defined by

$$L^R := \{ w^R \in \Sigma^* \mid w \in L \}.$$

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Let L be a language on alphabet Σ . The **reversal** of L is defined by

$$L^R := \{ w^R \in \Sigma^* \mid w \in L \}.$$

Theorem 4.11

If L is regular language, then so is L^R (proof using automata or regular expressions)

Closure Under Reversal

Proof using automata

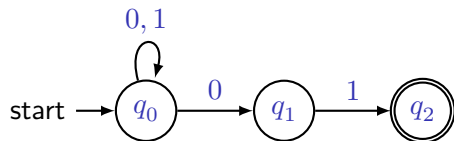
Let L be recognized by a finite automaton A . From the automaton A we get a finite automaton for L^R , by

1. Reversing all arcs.
2. Make the start state of A be the only accepting state.
3. Create a new start state p_0 with transitions $\delta(p_0, \varepsilon) = f$, where $f \in F$ are the accepting states of A . ■

Closure Under Reversal

Example

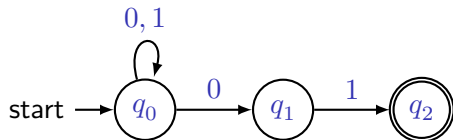
A NFA accepting all the binary strings that end in 01.



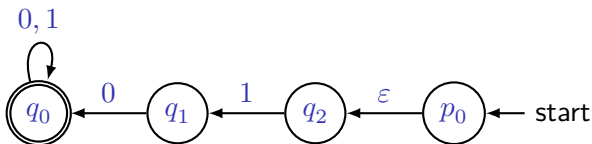
Closure Under Reversal

Example

A NFA accepting all the binary strings that end in 01.



A NFA accepting all the binary strings that start with 10.



Homomorphisms

Definition

An **algebraic structure** on a set $A \neq \emptyset$ is essentially a collection of n -ary operations on A (Birkhoff 1946, 1987).

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Example (Semigroup)

A **semigroup** $(S, *)$ is a set S with an associative binary operation $* : S \times S \rightarrow S$.

Homomorphisms

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Example (Semigroup)

A **semigroup** $(S, *)$ is a set S with an associative binary operation $* : S \times S \rightarrow S$.

Example (Monoid)

A **monoid** $(M, *, \varepsilon)$ is a semigroup $(M, *)$ with an element $\varepsilon \in M$ which is a unit for $*$, i.e. $(\forall x)(x * \varepsilon = \varepsilon * x = x)$.

Homomorphisms

Definition

A **homomorphism** is a structure-preserving map between two algebraic structures.

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Example

A homomorphism between two semigroups $(S, *)$ and $(S', *')$ is a function $\varphi : S \rightarrow S'$ such that:

$$(\forall x)(\forall y)[\varphi(x * y) = \varphi(x) *' \varphi(y)].$$

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$$(\forall x)(\forall y)[\varphi(x * y) = \varphi(x) *' \varphi(y)].$$

Graphically,

$$\begin{array}{ccc} (S, *) & \xrightarrow{x * y} & (S, *) \\ \downarrow \varphi(x), \varphi(y) & & \downarrow \varphi(x * y) \\ (S', *') & \xrightarrow{\varphi(x) *' \varphi(y)} & (S', *') \end{array}$$

Homomorphisms

Example

A homomorphism between two monoids $(M, *, \varepsilon)$ and $(M', *, \varepsilon')$ is a function $\varphi : M \rightarrow M'$ such that:

$$\begin{aligned} &(\forall x)(\forall y)[\varphi(x * y) = \varphi(x) *' \varphi(y)], \\ &\varphi(\varepsilon) = \varepsilon'. \end{aligned}$$

Homomorphisms

Definition

A homomorphism φ between two algebraic structures is (Cohn 1981):

- a **monomorphism** if φ is an injection,
- an **epimorphism** if φ is a surjection,
- an **endomorphism** if φ is from an algebraic structure to itself,
- an **isomorphism** if φ is a bijection,
- an **automorphism** if φ is a bijective endomorphism.

Closure Under Homomorphism

Definition

Let Σ and Γ be two alphabets. A **homomorphism** between (the monoids) $(\Sigma^*, \cdot, \varepsilon)$ and $(\Gamma^*, \cdot, \varepsilon)$ is a function

$$\begin{aligned}h &: \Sigma^* \rightarrow \Gamma^* \\a_1 a_2 \cdots a_n &\mapsto h(a_1) h(a_2) \cdots h(a_n) \\ \varepsilon &\mapsto \varepsilon\end{aligned}$$

Note: For this reason the textbook talks about a homomorphism $h : \Sigma \rightarrow \Gamma^*$.

Closure Under Homomorphism

Example

Let $h : \{0, 1\}^* \rightarrow \{a, b\}^*$ be a homomorphism defined by

$$h(0) = ab, \quad h(1) = \varepsilon.$$

Then

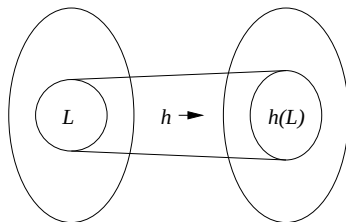
$$\begin{aligned} h(0011) &= h(0)h(0)h(1)h(1) \\ &= abab. \end{aligned}$$

Closure Under Homomorphism

Definition

Let L be a language over an alphabet Σ and let h be a homomorphism on Σ . The **application** of h to L , denoted $h(L)$, is defined by[†]

$$h(L) := \{ h(w) \mid w \in L \}.$$



[†]Figure from Hopcroft, Motwani and Ullman (2007, Fig. 4.5a).

Closure Under Homomorphism

Example

Let $h : \{0, 1\}^* \rightarrow \{a, b\}^*$ be a homomorphism defined by

$$h(0) = ab, \quad h(1) = \varepsilon.$$

If $L = L(\mathbf{10^*1})$, then $h(L) = L(\mathbf{(ab)^*})$.

Closure Under Homomorphism

Theorem 4.14

If L is a regular language over the alphabet Σ and h is a homomorphism on Σ , then $h(L)$ is also regular.

Closure Under Homomorphism

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If L is a regular language over the alphabet Σ and h is a homomorphism on Σ , then $h(L)$ is also regular.

Proof plan

- Let E be a regular expression such that $L = L(E)$.
- Let $h(E)$ be the regular expression replacing each symbol $a \in \Sigma$ by $h(a)$ in the regular expression E .
- We need to prove that $L(h(E)) = h(L(E))$.

(continued on next slide)

Closure Under Homomorphism

Proving $L(h(E)) = h(L(E))$

- Basis step
 - E is ε or $\emptyset$.
 1. $h(E) = E$ (h does not affect E)
 2. $h(L(E)) = L(E)$ ($L(E)$ is empty or only contains ε)
 3. $L(h(E)) = L(E) = h(L(E))$ (by 1 and 2)

(continued on next slide)

Closure Under Homomorphism

Proving $L(h(E)) = h(L(E))$

- Basis step
 - $E = a$
 1. $L(E) = \{a\}$
 2. $h(L(E)) = \{h(a)\}$
 3. $h(E)$ is the regular expression that is the string of symbols $h(a)$
 4. $L(h(E)) = \{h(a)\}$
 5. $L(h(E)) = h(L(E))$ (by transitivity between 2 and 4)

(continued on next slide)

Closure Under Homomorphism

Proving $L(h(E)) = h(L(E))$ (continuation)

- Inductive step
 - $E = F + G$
 1. $L(E) = L(F) \cup L(G)$ (def. of $+$)
 2. $h(E) = h(F + G) = h(F) + h(G)$ (def. of $h(E)$)
 3. $L(h(E)) = L(h(F) + h(G)) = L(h(F)) \cup L(h(G))$ (def. of $+$)
 4. $h(L(E)) = h(L(F) \cup L(G)) = h(L(F)) \cup h(L(G))$ (h is applied to a language by application to each of its strings)
 5. $L(h(F)) = h(L(F))$ and $L(h(G)) = h(L(G))$ (IH)
 6. $L(h(E)) = h(L(E))$

(continued on next slide)

Closure Under Homomorphism

Proving $L(h(E)) = h(L(E))$ (continuation)

- Inductive step
 - $E = FG$ (similar to the previous case)

Closure Under Homomorphism

Proving $L(h(E)) = h(L(E))$ (continuation)

- Inductive step
 - $E = FG$ (similar to the previous case)
 - $E = F^*$ (similar to the previous case)
 1. $L(E) = (L(F))^*$ (def. of $*$)
 2. $h(E) = h(F^*) = (h(F))^*$ (def. of $h(E)$)
 3. $L(h(E)) = L((h(F))^*) = (L(h(F)))^*$ (def. of $*$)
 4. $h(L(E)) = h((L(F))^*) = (h(L(F)))^*$ (h is applied to a language by application to each of its strings)
 5. $L(h(F)) = h(L(F))$ (IH)
 6. $L(h(E)) = h(L(E))$



Closure Under Homomorphism

Example

Let $\Sigma = \{0, 1, 2\}$. Prove that L is a language not regular.

$$L = \{ 0^i 1^j 2^k \mid i, j, k \in \mathbb{Z}^+ \text{ and } i \neq j \neq k \}.$$

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$$L = \{ 0^i 1^j 2^k \mid i, j, k \in \mathbb{Z}^+ \text{ and } i \neq j \neq k \}.$$

Proof

1. We define the homomorphism

$$h(0) = 0, \quad h(1) = 1, \quad h(2) = \varepsilon.$$

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Proof

1. We define the homomorphism

$$h(0) = 0, \quad h(1) = 1, \quad h(2) = \varepsilon.$$

2. The homomorphism h removes the 2^k s, so

$$h(L) = \{ 0^i 1^j \mid i, j \in \mathbb{Z}^+ \text{ and } i \neq j \}.$$

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$$h(L) = \{ 0^i 1^j \mid i, j \in \mathbb{Z}^+ \text{ and } i \neq j \}.$$

3. We know that $h(L)$ is not regular, so L is not regular. ■

Closure Under Homomorphism

Example

Let L be a regular language and h a homomorphism on L . Define $h^*(L)$ by

$$h^*(L) = L \cup h(L) \cup h(h(L)) \cup h(h(h(L))) \cup \dots$$

Is $h^*(L)$ necessarily regular?

[†]From somewhere in Internet (I don't remember).

Closure Under Homomorphism

Example

Let L be a regular language and h a homomorphism on L . Define $h^*(L)$ by

$$h^*(L) = L \cup h(L) \cup h(h(L)) \cup h(h(h(L))) \cup \dots$$

Is $h^*(L)$ necessarily regular?

Solution

No. Let $L = \{01\}$ and h defined as $h(0) = 00$ and $h(1) = 11$. Then

$$\begin{aligned} h^*(L) &= \{01, 0011, 00001111, \dots\} \\ &= \{0^n 1^n \mid n = 2^k \text{ for } k \geq 0\}, \end{aligned}$$

which is a language not regular.[†]

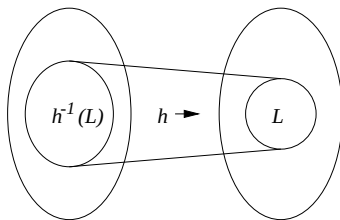
[†]From somewhere in Internet (I don't remember).

Closure Under Inverse Homomorphism

Definition

Let $h : \Sigma^* \rightarrow \Gamma^*$ be a homomorphism and $L \subseteq \Gamma^*$ a language. The **application** of h^{-1} to L , denoted $h^{-1}(L)$, is defined by[†]

$$h^{-1}(L) := \{ w \in \Sigma^* \mid h(w) \in L \}.$$



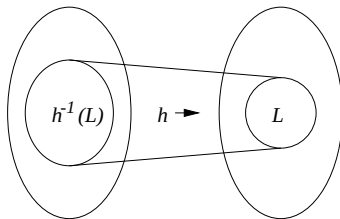
[†]Figure from Hopcroft, Motwani and Ullman (2007, Fig. 4.5b).

Closure Under Inverse Homomorphism

Definition

Let $h : \Sigma^* \rightarrow \Gamma^*$ be a homomorphism and $L \subseteq \Gamma^*$ a language. The **application** of h^{-1} to L , denoted $h^{-1}(L)$, is defined by[†]

$$h^{-1}(L) := \{ w \in \Sigma^* \mid h(w) \in L \}.$$



Observation

Note that h^{-1} is a relation but it is not necessarily a function.

[†]Figure from Hopcroft, Motwani and Ullman (2007, Fig. 4.5b).

Closure Under Inverse Homomorphism

Example

Let $h : \{a, b\} \rightarrow \{0, 1\}^*$ a homomorphism defined by

$$h(a) = 01, \quad h(b) = 10,$$

and let L be the language denoted by the regular expression $(\mathbf{00} + \mathbf{1})^*$, i.e.

$$L = \{ w \in \{0, 1\}^* \mid \text{all the } 0\text{'s occur in adjacent pairs} \}.$$

Closure Under Inverse Homomorphism

Example

Let $h : \{a, b\} \rightarrow \{0, 1\}^*$ a homomorphism defined by

$$h(a) = 01, \quad h(b) = 10,$$

and let L be the language denoted by the regular expression $(\mathbf{00} + \mathbf{1})^*$, i.e.

$$L = \{ w \in \{0, 1\}^* \mid \text{all the } 0\text{'s occur in adjacent pairs} \}.$$

Then

$$h^{-1}(L) = L((\mathbf{ba})^*).$$

Note that h^{-1} is not a function, but a relation.

It is necessary to prove $h(w) \in L \Leftrightarrow w = baba \cdots ba$.

Closure Under Inverse Homomorphism

Theorem 4.16

Let $h : \Sigma^* \rightarrow \Gamma^*$ be a homomorphism and $L \subseteq \Gamma^*$ a regular language. Then $h^{-1}(L)$ is regular (proof using automata).

Closure Under Inverse Homomorphism

Example

Prove that $L = \{ 0^n 1^{2n} \mid n \geq 0 \}$ is a language not regular.

Closure Under Inverse Homomorphism

Example

Prove that $L = \{ 0^n 1^{2n} \mid n \geq 0 \}$ is a language not regular.

Proof

1. Given the homomorphism

$$h(0) = 0, \quad h(1) = 11,$$

then

$$h^{-1}(L) = \{ 0^n 1^n \mid n \geq 0 \}.$$

Closure Under Inverse Homomorphism

Example

Prove that $L = \{ 0^n 1^{2n} \mid n \geq 0 \}$ is a language not regular.

Proof

1. Given the homomorphism

$$h(0) = 0, \quad h(1) = 11,$$

then

$$h^{-1}(L) = \{ 0^n 1^n \mid n \geq 0 \}.$$

2. Since $h^{-1}(L)$ is not regular, then L is not regular. ■

Some Exercises

Exercise 4.2.2

If L is a language, and a is a symbol, then L/a , the quotient of L and a , is the set of strings w such that wa is in L . For example, if $L = \{a, aab, baa\}$, then $L/a = \{\varepsilon, ba\}$. Prove that if L is regular, so is L/a . *Hint:* Start with a DFA for L and consider the set of accepting states.

Some Exercises

Exercise 4.2.2

If L is a language, and a is a symbol, then L/a , the quotient of L and a , is the set of strings w such that wa is in L . For example, if $L = \{a, aab, baa\}$, then $L/a = \{\varepsilon, ba\}$. Prove that if L is regular, so is L/a . *Hint:* Start with a DFA for L and consider the set of accepting states.

Proof (Hopcroft, Motwani and Ullman (2007) solution)

Start with a DFA A for L . Construct a new DFA B , that is exactly the same as A , except that state q is an accepting state of B if and only if $\delta(q, a)$ is an accepting state of A . Then B accepts input string w if and only if A accepts wa ; that is, $L(B) = L/a$. ■

Closure Properties

Exercise 4.2.3

If L is a language, and a is a symbol, then $a \setminus L$ is the set of strings w such that aw is in L . For example, if $L = \{a, aab, baa\}$, then $a \setminus L = \{\varepsilon, ab\}$. Prove that if L is regular, so is $a \setminus L$. *Hint:* Start with a DFA for L and consider its start state.

Closure Properties

Exercise 4.2.3

If L is a language, and a is a symbol, then $a \backslash L$ is the set of strings w such that aw is in L . For example, if $L = \{a, aab, baa\}$, then $a \backslash L = \{\varepsilon, ab\}$. Prove that if L is regular, so is $a \backslash L$. *Hint:* Start with a DFA for L and consider its start state.

Proof (Hopcroft, Motwani and Ullman (2007) solution)

Start with a DFA A for L . Construct a new DFA B , that is exactly the same as A , except that its start state is $\delta(q_0, a)$ where q_0 is the start state of A . Then B accepts input string w if and only if A accepts aw ; that is, $L(B) = L \backslash a$. ■

Closure Properties

Exercise 4.2.13.b

We can use closure properties to help prove certain languages are not regular. Start with the fact that the language

$$L_{0n1n} = \{ 0^n 1^n \mid n \geq 0 \}$$

is not a regular set. Prove that the following language not to be regular by transforming it, using operations known to preserve regularity, to L_{0n1n} :

$$L = \{ 0^n 1^m 2^{n-m} \mid n \geq m \geq 0 \}.$$

References



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— (1987). Universal Algebra. In: Selected Papers on Algebra and Topology by Garrett Birkhoff. Ed. by Gian-Carlo Rota and Joseph S. Oliveira. Birkhäuser, pp. 146–162 (cit. on pp. 26–28).



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John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on pp. 2, 9, 36, 51, 52, 59–62).